

August 7, 2026

Dear Hiring Team,

I am writing to apply for the Data Engineer I position at Finning International Inc. With a Master of Science in Computer Science from the University of Calgary and hands-on experience building Python/C++ software, SQL-backed backend services, ETL-style data pipelines, and scalable data-processing workflows, I am excited by the opportunity to contribute to Finning's data engineering team in Calgary or Edmonton.

My background aligns well with a role focused on building, supporting, and improving reliable data solutions. As a research software developer with BigGeo, Mitacs, and the University of Calgary, I built Python and C++ pipelines for ingesting, preprocessing, storing, retrieving, and evaluating large Earth observation datasets. I automated SAR and optical imagery preprocessing with GDAL, Rasterio, SNAP, and external data APIs, and developed DGGS-based workflows that transformed imagery into structured tensor files for downstream analysis.

That work required the same habits I would bring to Finning's data platforms: careful data handling, attention to data quality, performance awareness, and practical debugging. I improved one C++ processing workflow from 233 seconds to 26 seconds for a 10-tensor workload and reduced storage by 38% on a scaled Canada-wide dataset. I also worked with time-series data, multiresolution retrieval, and reproducible experimentation, which strengthened my ability to reason about pipeline design, reliability, and downstream analytical use.

I also bring backend and database experience from my work as a back-end developer. I developed Django REST APIs for NLP, text-to-speech, and speech-to-text services backed by PostgreSQL, implemented authentication flows, payment-service logic, and structured plan/payment records in a relational database. This experience gave me a practical foundation in SQL-backed systems, API integration, service reliability, and translating product requirements into maintainable software.

Alongside my development work, I supported University of Calgary courses in big data applications, working with data at scale, and database management systems. I mentored students on AWS-based data pipelines, cloud storage, debugging, scalable workflows, SQL, and database design. Those teaching responsibilities strengthened my communication skills and my ability to explain technical tradeoffs clearly to people with different levels of technical background.

I am especially interested in this role because it is an early-career data engineering opportunity close to Calgary, with a clear focus on practical pipelines, integration workflows, cloud and on-prem data systems, monitoring, troubleshooting, and supporting analytics. I have not worked directly with every enterprise tool in Finning's environment, such as Azure Data Factory or Databricks, but I have a strong base in Python, SQL, data pipelines, backend services, cloud coursework, and learning new systems quickly through hands-on work.

Thank you for considering my application. I would welcome the opportunity to discuss how my software engineering, data pipeline, SQL, and large-scale data-processing experience can contribute to Finning's data engineering team.

Sincerely,
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